

SEN.RTL - Technical Capabilities & IP Spec Sheet

Independent Digital Design & Semiconductor Engineering Portfolio

Email: design@sen-rtl.com | Repository: github.com/sen-rtl

Domain Focus: RTL Core Design, FPGA IP, VLSI Verification & STA

PLATFORM OVERVIEW

SEN.RTL is an engineering platform demonstrating advanced digital logic design, RTL modeling, FPGA prototyping, and synthesis-aware VLSI design flows.

IP CORE CATALOG & FEATURES

1. RISC-V Pipelined ALU
 - 4-stage pipelined execution datapath matching RV32I specification.
 - Resolves RAW data hazards via bypass forwarding logic and stalls.
2. Parameterized N-Bit Barrel Shifter
 - Cascaded 2-to-1 multiplexer tree layout ($\log_2 N$ depth).
 - Single-cycle shifting & rotation (LSL, LSR, ASR) with zero inferred latches.
3. Systolic Matrix Multiplier Core
 - 4x4 spatial array of processing element (PE) MAC units.
 - High-throughput parallel arithmetic optimized for tensor operations.

TECHNICAL VERIFICATIONS

- Verilog HDL & SystemVerilog functional testbenches and simulations.
- Static Timing Analysis (STA) setup/hold constraints and clock gating.
- Synthesis mapping, floorplanning, and routing closure in ASIC flows.